

Area of parallelogram and triangle

Unit 2 · area · 65 minutes · 1-to-3 Online Lesson

Corresponding textbooks: "New Thinking in Primary School Mathematics (Second Edition)" Volume 5, Volume A, Unit 2 + Modern Education, Volume 5, Unit 5

Core Trap: |||SEP||| ● High frequency

The sub-test is compulsory every year, and the area question accounts for about 10-15% of Paper 1

Prerequisite knowledge: P4 area of rectangle/square, P4 concepts of vertical and parallel lines, P4 simple graphic segmentation

Course goal: ❶ Master the formula for the area of a parallelogram ❷ Master the formula for the area of a triangle ❸ Reverse the height/base of a given area ❹ Comparison and application of areas

Our goals: ❶ Master the formula for the area of a parallelogram ❷ Master the formula for the area of a triangle ❸ Find the height/base of a given area ❹ Comparison and application of areas

Student Name:

Class:

Date:

Time Spent:

I.Warm-Up Questions (total 5 question , 5 minutes)

#	Question	Difficulty	Working Space (Show full working)
1	Calculate the area of the rectangle: length 8 cm, width 5 cm.	Basic	
2	Calculate the area of the square: side length 6 cm.	Basic	
3	On graph paper, from A to B, move 4 squares to the right and 3 squares up. Is the straight line distance between AB hypotenuse or perpendicular?	Basic	
4	A parallelogram has a base of 10 cm but has a marked hypotenuse of 6 cm. Is this hypotenuse high? Why?	Advanced	
5	How many bases does a triangle have? How many highs does each bottom correspond to? (Tip: Either side can be the bottom)	Advanced	

II.Core Knowledge + Worked Examples

Knowledge point 1: Area of parallelogram = base $\times$ height (height is the vertical distance!) ●

SSPA

- ① **Formula for the area of a parallelogram** : Area = Base $\times$ Height
- ② **Height |||SEP|||: Measure vertically (90°) upward from the base to the opposite side** The vertical distance **is not the length of the hypotenuse!** ||| SEP || , **Not the length of the hypotenuse!**
- ③ **end:** Any side can be the base, but the height must be the vertical distance corresponding to that base.
- ④ Parallelograms can be "cut and complemented" into rectangles (cut triangles are translated) $\rightarrow$ so the formula is the same as that of rectangles
- ⑤ Unit: The area unit is the square unit (cm^2 , m^2 , km^2)

SVG: Parallelogram (base and height)



Trap detonation example (the most important demonstration in this class)

The parallelogram has a base of 10 cm, a hypotenuse of 8 cm, and a height of 6 cm. Find the area.

✗ Common mistakes (70% students)

$$10 \times 8 = 80 \text{ cm}^2$$

Use the hypotenuse as the height! The hypotenuse is not a vertical distance!

✓ Correct solution

$$10 \times 6 = 60 \text{ cm}^2$$

Height = vertical distance = 6 cm (not hypotenuse 8 cm!)

Tip: "The base times the height, the height is vertical; the hypotenuse is not the height, the height must be measured at a right angle!"

⚠ The most frequent error: using the hypotenuse of a parallelogram as the height! The height must be the vertical ($\perp$) distance from the bottom edge to the opposite edge.

⚠ The second most common mistake: forgetting to write the area unit cm^2 . The area must have the word "square"!

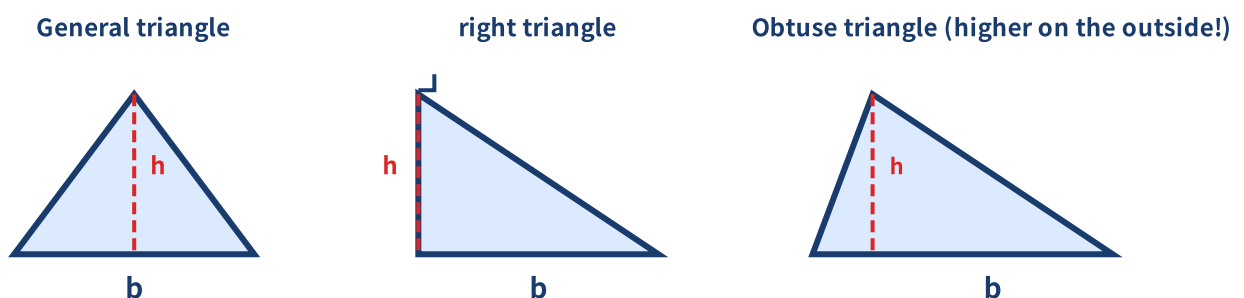
Knowledge Point - Worked Examples

#	Question	Difficulty	Working Space
Example 1	The parallelogram has a base of 12 cm and a height of 5 cm. Find the area.		
Example 2	The parallelogram has a base of 8 m and a height of 3.5 m. Find the area.		

Knowledge point 2: Area of a triangle = base $\times$ height $\div$ 2 (the same for any triangle!) SSPA required test

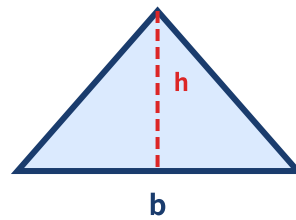
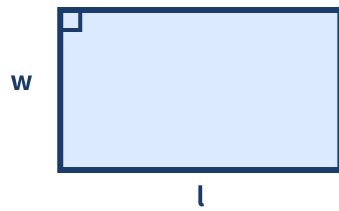
- ① **Triangle area formula** : Area = base $\times$ height $\div$ 2
- ② **Why $\div$ 2?** Two identical triangles put together = a parallelogram, so the area of the triangle is half of the parallelogram. Any triangle will do |||SEP|||: right triangle, isosceles triangle, obtuse triangle - the formulas are the same!
- ③ **The height must be perpendicular to the base:** Use a ruler/protractor to confirm 90°
- ④ **The height of an obtuse triangle is outside:** When the opposite angle to the base is an obtuse angle ($>90^\circ$), the height will fall on the extension line of the base |||SEP|||: When the opposite angle to the base is an obtuse angle ($>90^\circ$), the height will fall on the base extension cord superior

SVG: Triangle (base and height)



SVG: Parallelogram vs triangle area comparison

$$\text{Area of parallelogram} = \text{base} \times \text{height} \quad \text{Area of triangle} = \text{base} \times \text{height} / 2$$



The area of a parallelogram is 2 times the area of a triangle! When the base is the same and the height is the same, the area of a triangle is half that of a parallelogram.

Trap detonation example

Triangular base 10 cm, height 6 cm. Find the area.

✗ Common mistakes (50% students)

$$10 \times 6 = 60 \text{ cm}^2$$

Forget $\div 2$! Directly used the parallelogram formula!

✓ Correct solution

$$10 \times 6 \div 2 = 30 \text{ cm}^2$$

Area of triangle = base $\times$ height $\div 2$, be sure to remember $\div 2$!

⚠ The most frequent error: forgetting the area of a triangle $\div 2$! Two identical triangles equal a parallelogram!

⚠ Obtuse triangle trap: The height is outside the triangle (on the extension of the base), and students cannot find the height!

Knowledge point 3: Find the high or bottom of the known area ● SSPA Advanced

Reverse formula (algebraic thinking):

① **Parallelogram:** Height = Area $\div$ Base | Base = Area $\div$ Height

② **Triangle:** Height = Area $\times 2 \div$ Base | Base = Area $\times 2 \div$ Height

Change it back to a parallelogram, and then $\div$ the base to find the height)

③ Common test results: given the area and base, find the height (or give the area and height, find the base)

1

parallelogram

Height=Area $\div$ Base=Area $\div$ Height

2

triangle

Height=Area $\times 2 \div$ Base=Area $\times 2 \div$ Height

3

First $\times 2$

Remember to multiply the triangle by 2 first (replace half)

4

Units are consistent

Area cm^2 , base cm and height cm, the units are unified

example

Example 3: The area of a parallelogram is 48 cm^2 and the base is 8 cm. Seek high.

example

Example 4: The area of a triangle is 24 cm^2 , and the base is 8 cm. Seek high. (Note: $\times 2$ first!)

Knowledge Point 1 ~ 3 Worked Examples (write out formula $\rightarrow$ substitute $\rightarrow$ answer with units)

#	Question	Difficulty	Working Space
Example 5	Triangular base 15 cm, height 8 cm. Find the area.		
Example 6	The area of the triangle is 36 cm^2 and the base is 9 cm. Seek high.		
Example 7	The corresponding bases and heights of the two sets of parallelograms are: base $_1=10$ cm, height $_1=4$ cm; base $_2=5$ cm. Ask for a higher $_2$.		
Example 8	The triangle has an area of 20 cm^2 and a height of 5 cm. Seek bottom.		

Knowledge point 4: Area comparison and application SSPA application

- ① Parallelograms and triangles with the same base and the same height $\rightarrow$ The area of a parallelogram is twice that of a triangle **2 times**
- ② When comparing the areas of figures: first calculate the areas separately $\rightarrow$ then compare $\rightarrow$ finally express with "more/less" or "multiple"
- ③ Combined figures: Divide into basic figures (rectangle + triangle) $\rightarrow$ Calculate separately $\rightarrow$ Add/subtract

III. Lesson Layered Synchronization Practice

Basic layer (total 7 questions, everyone must do)

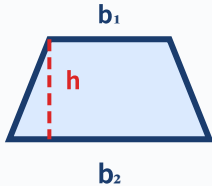
#	Question	Difficulty	Working Space
1	The parallelogram has a base of 9 cm and a height of 6 cm. Find the area.		
2	Triangular base 10 cm, height 4 cm. Find the area.		
3	The parallelogram has a base of 7 m and a height of 5 m. Find the area.		
4	Triangular base 12 cm, height 5 cm. Find the area.		
5	The area of a parallelogram is 72 cm^2 and its base is 8 cm. Seek high.		

6	Triangular base 6 cm, height 8 cm. Find the area.		
7	The parallelogram has a base of 15 m and a height of 4 m. Find the area.		

Advanced layer (total 5 questions, choose do)

#	Question	Difficulty	Working Space
8	The area of a triangle is 40 cm^2 and its base is 10 cm. Seek high.		
9	A parallelogram has an area of 96 cm^2 and a height of 8 cm. Seek bottom.		
10	A parallelogram and a triangle have the same base (12 cm) and height (7 cm). How many times the area of a parallelogram is that of a triangle?		
11	The triangle has an area of 54 cm^2 and a height of 9 cm. Seek bottom.		
12	A parallelogram has a base of 20 cm and an area of 160 cm^2 . Seek high.		

 challenge layer (total 4 questions, choose do, SSPAKiller Questions)

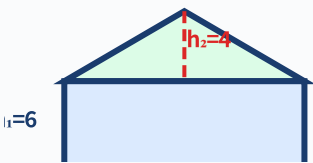
#	Question	Difficulty	Working Space
13	An obtuse triangle with base 14 cm. The vertical distance (height) from the vertex to the bottom extension line is 6 cm. Find the area of the triangle.		
14	Two sets of bases and heights of a parallelogram: $\text{base}_1=10 \text{ cm}$, $\text{height}_1=6 \text{ cm}$; $\text{base}_2=8 \text{ cm}$, $\text{height}_2=?$ Seek high SEP . (Hint: The area of the same parallelogram is fixed) ₂ . (Hint: The area of the same parallelogram is fixed)		
15	 A trapezoid can be divided into a parallelogram and a triangle. The parallelogram has a base of 8 cm and a height of 5 cm; a triangle has a base of 4 cm and a height of 5 cm. Find the area of the entire trapezoid.		

16	Two different triangles A and B. A: Bottom 10 cm, height 6 cm. B: Bottom 15 cm, height 4 cm. Compare the areas of A and B, which one is larger? How much bigger?		
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areaapplicationquestion (all must do, column → formula → calculate → answer sentence)

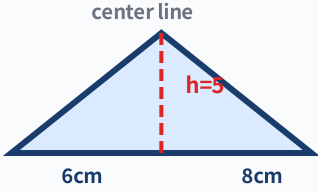
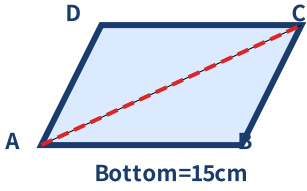
#	Question	Difficulty	Working Space
17	A parallelogram flowerbed with a base of 15 m and a height of 8 m. Find the area of the flower garden.		
18	A triangular piece of grass with a base of 20 m and a height of 12 m. If 3 flowers are planted per square meter, how many flowers can be planted on the lawn?		
19	Parallelogram rug area 6 m^2 , base 2.5 m. Seek high.		
20	The triangular billboard has an area of 3 m^2 and a height of 1.5 m. Find the bottom length.		

advancedapplicationquestion (  choose do, SSPAPaper 2commonquestion type)

#	Question	Difficulty	Working Space
21	 The picture below is made up of a rectangle and a triangle. The rectangle is 10 cm long and 6 cm wide; the triangle has a base of 10 cm and a height of 4 cm (the same base as the rectangle). Find the area of the whole picture.		
22	Parallelogram A: base 12 cm, height 5 cm. Triangle B: base 10 cm, height 6 cm. Which shape has the larger area? How much bigger?		

IV. Ultimate challenge area

#	Question	Difficulty	Working Space
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 1	 A large triangle is divided into two smaller triangles by a midline. The bases of the two small triangles are 6 cm and 8 cm respectively, and the heights are both 5 cm. (a) What are the areas of each of the two small triangles? (b) What is the area of the large triangle? (c) Verify: The area of the large triangle = the sum of the areas of the two small triangles.	
 2	 As shown in the figure, parallelogram ABCD has base AB = 15 cm and height = 8 cm. Diagonal AC divides the parallelogram into two triangles. (a) What is the area of parallelogram ABCD? (b) What is the area of triangle ABC? (c) How can you derive the area of a triangle from the area of a parallelogram without using a formula?	
 3	A combined figure is formed by overlapping a parallelogram and a triangle. A parallelogram has a base of 20 cm and a height of 10 cm; a triangle has a base of 12 cm and a height of 8 cm (the triangle is completely within the parallelogram). Question: (a) What is the area of the parallelogram? (b) What is the area of the triangle? (c) What is the area of the part of the parallelogram not covered by the triangle?	

V. Class afterhomework

Basic must-do questions (total 6 questions, must write formula and step)

#	Question	Difficulty	Working Space
H1	The parallelogram has a base of 11 cm and a height of 7 cm. Find the area.		
H2	Triangular base 8 cm, height 9 cm. Find the area.		
H3	The area of a parallelogram is 84 cm^2 and its base is 12 cm. Seek high.		
H4	The area of the triangle is 30 cm^2 and the base is 6 cm. Seek high. (Note: $\times 2$ first!)		

H5	Parallelogram floor tiles have a base of 30 cm and a height of 20 cm. Find the area of the floor tiles.		
H6	Triangular flag base 60 cm, height 40 cm. Find the area of the flag.		

For advanced, choose doquestion (total 2 questions,  choose do)

#	Question	Difficulty	Working Space
H7	The parallelogram and the triangle have the same base and height, the base is 16 cm and the height is 10 cm. (a) Find the areas of the two figures respectively. (b) How many times the area of a parallelogram is the area of a triangle?		
H8	Obtuse triangle with longest side 25 cm. Taking the longest side as the base, the vertical distance from the vertex to the extended line of the base is 12 cm. Find the area of the triangle. (Hint: the height is outside the triangle, but the formula remains the same!)		

VI. The Lessoncorecommon errorssummary

☒ Self-examination in this hall (tick after completion)

☐ I know the pitfalls of solving each knowledge point

☐ I can complete  basic questions independently

☐ I can challenge  advanced questions

☐ I remember the formula

Review of Learning Objectives - After completing this lesson you should be able to:

☐ Identify all trap types in our hall

☐ Solve  basic questions independently (100% correct)

☐ Challenge  Advanced questions (80%+ correct)

☐ Explain the lesson formula to classmates

#	common error	Correct Approach
1	Parallelogram: treat the hypotenuse as height SEP : base × hypotenuse : Bottom × hypotenuse	Height = vertical distance (⊥) from the bottom side to the opposite side, not the length of the hypotenuse!
2	Triangle: Forget ÷ 2 SEP : Base × height when the answer : Bottom × high when the answer	Area of triangle = base × height ÷ 2 SEP , two triangles are a parallelogram , two triangles are a parallelogram
3	Obtuse triangle: height not found	High above the base extension line (outside the triangle), draw a vertical line from the vertex to the base extension line

4	When you know the area and calculate the height/base, forget that the triangle must be $\times 2$ first	Triangle: Height = Area $\times 2 \div$ Base (not Area $\div$ Base!)
5	Forgot to write the area unit SEP : The answer is "60": Write "60" as the answer	The area unit must have "square": cm ² , m ² , km ²
6	Bottom and height do not correspond SEP : Use bottom A but measure the height corresponding to bottom B: Use base A but measure the corresponding height of base B	Each bottom has its own corresponding height and must be used in pairs!

Lam Fung Academy • LF Academy • We don't teach math. We teach trap avoidance.

 Related topics: L03 Area of parallelograms and triangles • L04 Area of trapezoidal polygons • L05 Special area trap